AUTOSAR HIGH-LEVEL DESIGN (HLD) SPECIFICATION
Document ID: HLD-POWERTRAIN-2026-V4.2
AUTOSAR Release: Classic Platform 4.4.0
Target ECU: EngineControlUnit_Main (ECU_Engine), BatteryManagementSystem (BMS_ECU)

1. Executive Summary & System Overview
This High-Level Design document specifies the software architecture for the Electric Vehicle Powertrain Control System (EVPCS) conforming to the AUTOSAR Classic 4.4 architecture standard. The system is responsible for torque demand arbitration, regenerative braking coordination, and thermal safety monitoring across the dual-inverter powertrain.

2. Software Architecture Scope
The EVPCS consists of multiple AUTOSAR Application Software Components (SWCs), Sensor-Actuator Software Components, and interactions with the Basic Software (BSW) stack via the Virtual Functional Bus (VFB) and Run-Time Environment (RTE).

Document Version: 4.2.1
Status: Approved for Architecture Review
Author: Powertrain Systems Architecture Team

3. Software Components (SWC) Inventory

3.1 TorqueArbitration_SWC
- Type: Application Software Component (SWC)
- Purpose: Arbitrates torque requests from AcceleratorPedal_Sensor, CruiseControl_SWC, and StabilityControl_SWC.
- Source Page Hint: Section 3.1, Page 2
- Ports:
  * R-Port: R_AccelPedalPos (Interface: If_PedalPosition)
  * R-Port: R_CruiseTorqueReq (Interface: If_TorqueRequest)
  * P-Port: P_FinalTorqueCommand (Interface: If_TorqueCommand)

3.2 BatteryThermalMonitor_SWC
- Type: Sensor-Actuator Software Component
- Purpose: Monitors battery module cell temperatures, calculates cooling pump request states, and flags thermal derating.
- Ports:
  * R-Port: R_BatteryTempSensor (Interface: If_RawTemperature)
  * P-Port: P_ThermalDeratingFactor (Interface: If_DeratingStatus)
  * P-Port: P_CoolantPumpCommand (Interface: If_PumpActuation)

3.3 InverterMotorControl_SWC
- Type: Complex Device Driver / Application SWC
- Purpose: Converts target torque setpoints into 3-phase PWM gating commands via the microcontroller hardware timer units.
- Dependencies: Requires P_FinalTorqueCommand from TorqueArbitration_SWC.

4. Software Interfaces & Communication Signals

4.1 AUTOSAR Interfaces
- Interface: If_PedalPosition
  * Type: SenderReceiver
  * Description: Carries normalized driver pedal demand (0.0 to 100.0%).
- Interface: If_TorqueCommand
  * Type: SenderReceiver
  * Description: Carries signed target torque in Newton-meters (Nm) to inverter.
- Interface: If_ThermalDeratingFactor
  * Type: SenderReceiver
  * Description: Transmits engine power limits based on pack temperature.
- Interface: If_DiagnosticsService
  * Type: ClientServer
  * Description: Diagnostic fault recording interface to Dem (Diagnostic Event Manager).

4.2 Signals & Network Data Elements
- Signal: Sig_DriverTorqueDemand
  * Data Type: float32
  * Direction: Inward (from Pedal Sensor)
  * Description: Driver pedal position sampled at 10ms periodic task.
- Signal: Sig_InverterTorqueCmd
  * Data Type: int16
  * Direction: Outward (to Inverter Controller)
  * Description: Filtered and slew-rate limited torque demand transmitted over CAN-FD bus.
- Signal: Sig_PackMaxTemp
  * Data Type: uint8
  * Direction: Inward
  * Description: Maximum pack temperature in degrees Celsius (-40 to +120 C).

5. Functional Execution Flows

5.1 Torque Delivery Sequence Flow
Flow Sequence:
Step 1: Driver presses accelerator pedal; AcceleratorPedal_Sensor reads dual hall-effect analog voltage channels.
Step 2: RTE passes validated sensor reading to TorqueArbitration_SWC via R_AccelPedalPos.
Step 3: TorqueArbitration_SWC checks BatteryThermalMonitor_SWC for thermal derating limits.
Step 4: Arbitrated torque command is dispatched via P_FinalTorqueCommand to RTE.
Step 5: RTE routes signal across Com/CanTp stack to InverterMotorControl_SWC on the powertrain CAN-FD network.

6. Architectural Dependencies and Known Limitations
- TorqueArbitration_SWC requires a 10ms cyclic trigger from OsTask_Powertrain_10ms.
- BatteryThermalMonitor_SWC requires NvM (Non-Volatile Memory) block allocation for calibration storage.
- Notice: The connection between StabilityControl_SWC and TorqueArbitration_SWC is referenced in the architecture overview but the interface port definition R_ESP_Intervention is currently missing from Section 4.
- Incomplete Specification: Emergency shutdown flow under DC link overvoltage does not explicitly detail the handshake timeout with the High Voltage Interlock Loop (HVIL).